港式拼音

HK Pinyin for Ubuntu

A macOS-style Traditional Chinese input method with Hanyu Pinyin, Hong Kong glyphs, vocabulary, punctuation and date formats.

Version 1.0.0 • fcitx5 + Rime • tested on Ubuntu 24.04 LTS • 19 October 2026



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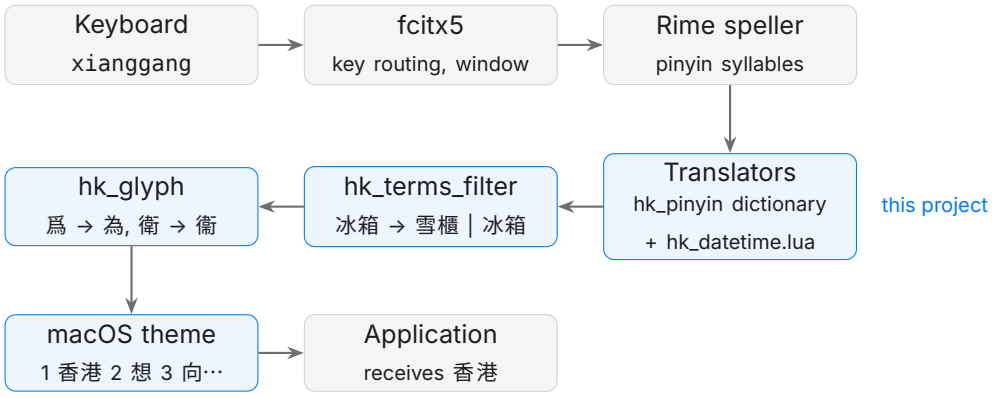
1 What it is

港式拼音 is an input method for Ubuntu that behaves like the *Pinyin –Traditional* input source on macOS, but writes the way Hong Kong writes. You type Hanyu Pinyin without tones; it gives Traditional Chinese in a single horizontal candidate row.

Input	Hanyu Pinyin, full syllables or initials (`xg` → 香港), whole-sentence conversion, learns the words you choose.
Characters	Hong Kong forms: 為衞線枱啟群眾, keeping the Big5 code points used on HK websites (戶說稅).
Vocabulary	Hong Kong terms first: 軟件網絡的士雪櫃電郵甚麼. 376 built-in HK words, place names and written-Cantonese words (嘅咗佢哋唔該).
Punctuation	「」『』、，。……—— 《》, all on the keys a macOS user expects.
Dates	`rq` 2026年10月19日 • 19/10/2026; `sj` 下午3:05; `xq` 星期一.
Look & keys	macOS light and dark candidate windows; `Ctrl+Space` switches input source; `Shift` toggles English.

1.1 How it is built

There is no new C code in the typing path. 港式拼音 is a Rime *schema* running inside fcitx5, the standard input-method framework on Ubuntu, so it gets fcitx5's support for GTK, Qt, X11 and Wayland applications for free.



The dictionary `hk_pinyin` is Rime's 朙月拼音 (`luna_pinyin`) word list, plus the Hong Kong word list, plus your own words. Two filters then rewrite each candidate: the first puts the Hong Kong term ahead of the mainland or Taiwan term, the second changes inherited character forms to the Hong Kong ones.

2 Install

2.1 Requirements

- Ubuntu 24.04 LTS (tested) or newer, or Debian 12+. Ubuntu 22.04 works with a fallback: without `librime-plugin-lua` you lose the `rq`/`sj` date shortcuts, and Hong Kong terms replace the original instead of being listed first.
- GNOME, KDE Plasma, Xfce or any other desktop. Both X11 and Wayland.
- About 60 MB of packages if fcitx5 is not already installed.

2.2 Steps

Download the project and run the installer *as yourself* (it calls `sudo` only for `apt`):

```
git clone https://github.com/YOUR-USERNAME/hk-pinyin-ime.git
cd hk-pinyin-ime
./install.sh
```

The installer:

1. installs fcitx5, fcitx5-rime, the luna_pinyin data, librime-lua and the Noto CJK HK fonts;
2. copies the schema, dictionaries, Lua scripts and OpenCC tables to `~/.local/share/fcitx5/rime/` and compiles them;
3. installs the `macos-light` and `macos-dark` themes;
4. sets up fcitx5: English + 港式拼音, `Ctrl+Space` to switch, horizontal candidates, Noto Sans CJK HK, input state shared across windows;
5. makes fcitx5 the system input method (`im-config`, `environment.d`, autostart).

Any fcitx5 settings it replaces are copied to `~/.config/fcitx5/hk-pinyin-backup-<date>/` first.

Then log out and log back in once.

> **GNOME on Wayland** Install the GNOME Shell extension *Input Method Panel* (kimpanel) from `extensions.gnome.org`. Without it the candidate window may appear at a corner of the screen instead of under the cursor in some applications.

2.3 Options

`--no-apt`	Do not install packages (you already have fcitx5 and fcitx5-rime).
`--layout gb`	Keyboard layout for the English input source. Default: detected, otherwise `us`.
`--no-config`	Only install the schema and themes; keep your fcitx5 settings. Add 港式拼音 yourself in *Fcitx 5 Configuration*.

3 Using it

3.1 Keys

Key	港式拼音 on Ubuntu	macOS equivalent
`Ctrl+Space`	Switch English ⇄ 港式拼音	`Ctrl+Space`
`Shift`	Toggle English inside 港式拼音; letters already typed are entered as English	`Caps Lock` / `Shift`
`Space` `1`–`9`	Choose a candidate	same
`-` `=` `Tab`	Previous / next page	same
`←` `→`	Move through candidates	same
`Enter`	Enter the letters as typed	same
`Esc`	Cancel	same
`Ctrl+`` or `F4`	Options menu (below)	—
`Shift+Space`	Half-width ⇄ full-width	—
`Ctrl+.`	Chinese ⇄ English punctuation	—

Type a capital letter first (`Hello`) to write English directly without switching. Type `/` and a code for symbols: `/fh` © ® ™, `/jt` arrows, `/sx` maths.

3.2 Options menu

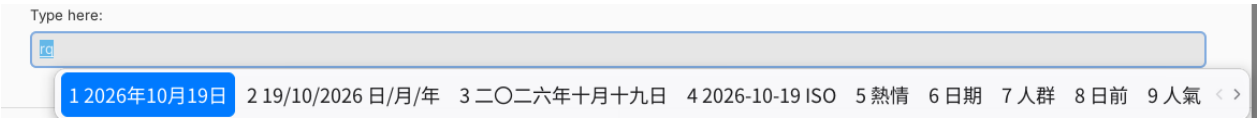
`Ctrl+`` opens the options. Your choices are remembered.

Option	On (default)	Off
香港詞彙	雪櫃 first, 冰箱 second	冰箱 only
香港字形	為衛線	爲衛綫 (Rime's inherited forms)
全形 / 半形	—	half-width (default)
中文標點	「，。」	ASCII punctuation

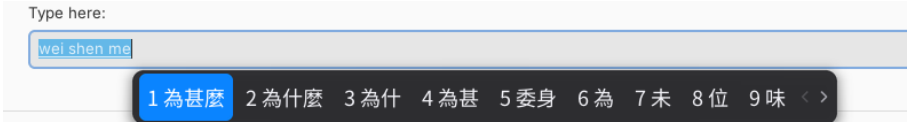
3.3 What it looks like



Hong Kong term first, the mainland/Taiwan word right after: `bingxiang` → 1 雪櫃 2 冰箱



rq gives today's date in Hong Kong formats



The dark theme follows the desktop's dark mode: `weishenme` → 為甚麼

4 Hong Kong conventions

4.1 Characters

Rime's dictionary stores inherited (舊字形) forms. The `hk_glyph` filter changes them to the forms in everyday Hong Kong use —the ones on government websites, in newspapers and produced by Big5/HKSCS input methods such as 倉頡 and 速成.

爲→為	衛→衛	綫→線	檯→枱	啓→啟	羣→群	衆→眾	裡→裏
牀→床	峯→峰	敘→敍	鉤→鈎	蔥→葱	菸→煙	卧→臥	麪→麵

Characters such as 戶說稅脫溫 are deliberately *not* changed to 户説税脱温. Those are the shapes in the 常用字字形表 reference, but Hong Kong text uses the Big5 code points and lets the HK font draw the local shape. If you want the strict reference forms, see §5.3.

4.2 Vocabulary

Hong Kong words are in the dictionary with high weights, and a filter rewrites mainland or Taiwan terms inside any candidate, sentence or phrase, while keeping the original as the next choice.

You type	First	Instead of	You type	First	Instead of
ruanjian	軟件	軟體	dishi	的士	計程車
wangluo	網絡	網路	bashi	巴士	公車
dayinji	打印機	印表機	bingxiang	雪櫃	冰箱
shujuku	數據庫	資料庫	lianjie	連結	鏈接
dianyou	電郵	電子郵件	zhuguli	朱古力	巧克力
duanxun	短訊	簡訊 / 短信	xuegao	雪糕	冰淇淋
shangzai	上載	上傳	sanwenzhi	三文治	三明治
shuma	數碼	數位	danche	單車	腳踏車
rengongzhineng	人工智能	人工智慧	jiemian	介面	界面
shenme	甚麼	什麼	anjie	按揭	房貸

The built-in list also covers government bodies (立法會, 廉政公署, 食環署, 強積金), housing (公屋, 居屋, 劏房), transport (港鐵, 八達通, 小巴), about 70 districts (旺角, 尖沙咀, 鰂魚涌, 赤鱲角) and common written-Cantonese characters and words (嘅咗喺唔冇佢哋嘢・唔該我哋而家點解搞掂). For these, a sensible Mandarin reading is used: 佢 `qu`, 嘅 `ge`, 冇 `mao`, 喺 `xi`, 嘢 `ye`, 哋 `di`.

Where a Hong Kong word shares its pinyin with a very common word, the common word stays first (`zhishi` → 只是, then 芝士). After you choose 芝士 a few times, it moves up for you.

4.3 Punctuation

Key	Gives	Key	Gives	Key	Gives	Key	Gives
`,`	，	`[`	「	`"`	「then 」	`\`	、
`.`	。	`]`	」	`'`	『then 』	`^`	……
`?`	？	`{`	『	`<`	《〈«	`_`	——
`!`	！	`}`	』	`>`	》〉»	`$`	$ HK$ ¥ € £
`;`	；	`(`	（	`:`	：	`*`	* × ※

Keys with several choices (`<`, `$`, `*`) show a small menu; press the key again or a number.

4.4 Dates and times

`rq`	2026年10月19日・19/10/2026 ・二〇二六年十月十九日・2026-10-19
`sj`	下午3:05 ・15:05 ・下午三時零五分 (三時正 on the hour)
`xq`	星期一・週一・(一)
`dt`	2026年10月19日（星期一）下午3:05 ・19/10/2026 15:05

Numeric dates are day/month/year, as in Hong Kong. Periods are 凌晨, 上午, 下午, 晚上.

5 Customising

All customisation lives in `~/.local/share/fcitx5/rime/`. After a change, choose *Deploy* (重新部署) in the fcitx5 tray menu, or run `fcitx5 -r`. Reinstalling never overwrites your own files:

- `hk_pinyin.user_terms.dict.yaml` —your words;
- `hk_pinyin.custom.yaml` —your settings (only on systems without librime-lua does the installer write it, after making a backup).

5.1 Add your own words

Edit `hk_pinyin.user_terms.dict.yaml`. Below the `...` line, one word per line, separated by *Tab* characters: word, pinyin, weight.

```
長洲太平清醮        chang zhou tai ping qing jiao        50000
將軍澳中心          jiang jun ao zhong xin          50000
```

Use `v` for ü (`lv` 律, `nv` 女). A weight of 50000 puts a word ahead of most homophones.

5.2 Common changes

Create `hk_pinyin.custom.yaml` with any of these under `patch:`.

```
patch:
  # fewer candidates per page
  menu/page_size: 6

  # fuzzy pinyin: z=zh c=ch s=sh, an=ang en=eng in=ing
  speller/algebra/+:
    - derive/^([zcs])h/$1/
    - derive/^([zcs])([^h])/$1h$2/
    - derive/([aei])n$/$1ng/
    - derive/([aei])ng$/$1n/

  # start each window in English instead of Chinese
  switches/@0/reset: 1
```

5.3 Strict 常用字字形表 character forms

To get 户説税温 and the other reference shapes, copy OpenCC's `t2hk` tables and point the glyph filter at them:

```
cp /usr/share/opencc/t2hk.json /usr/share/opencc/HKVariants.ocd2 \
   ~/.local/share/fcitx5/rime/opencc/
```

```
patch:
  hk_glyph/opencc_config: t2hk.json
```

5.4 Window style

Fcitx 5 Configuration → Addons → Classic User Interface: choose `macos-light` / `macos-dark`, the font, or a vertical list. To change colours, edit `THEMES` in `tools/make_themes.py`, run it, and run `./install.sh --no-apt` again.

6 Project files

`install.sh, uninstall.sh`	Installer and remover.
`rime/hk_pinyin.schema.yaml`	The schema: switches, engine pipeline, punctuation, keys.
`rime/hk_pinyin.dict.yaml`	Main dictionary: imports luna_pinyin + HK terms + user terms.
`rime/hk_pinyin.hk_terms.dict.yaml`	376 Hong Kong words (generated).
`rime/hk_pinyin.user_terms.dict.yaml`	Your words (installed once, never overwritten).
`rime/default.custom.yaml`	Global Rime settings: page size, Shift behaviour, menu keys.
`rime/lua/hk_datetime.lua`	`rq sj xq dt`.
`rime/lua/hk_terms_filter.lua`	Puts Hong Kong terms first, keeps the original next.
`rime/opencc/hk_terms.txt`	Mainland/Taiwan → Hong Kong term pairs.
`rime/opencc/hk_glyph.txt`	Inherited → Hong Kong character forms.
`themes/macos-light, macos-dark`	fcitx5 themes (generated).
`tools/hk_terms.src, build_dict.py`	Source of the HK word list and its generator (checks syllable counts).
`tools/make_themes.py`	Draws the theme images and writes `theme.conf`.
`tests/rime_probe.c, run_tests.sh`	Types pinyin into librime and checks 26 expected results.
`docs/`	This manual (`.tex` and `.pdf`).

6.1 Development loop

```
sudo apt install gcc librime-dev librime-plugin-lua rime-data-luna-pinyin python3-pil
vim tools/hk_terms.src            # add words:  詞語 | pin yin | weight
python3 tools/build_dict.py       # regenerate the dictionary
tests/run_tests.sh                # 26 checks, no desktop needed
./install.sh --no-apt             # try it for real
```

7 Troubleshooting

Problem	Fix
Ctrl+Space does nothing	Log out and in after installing. Check `im-config -m` shows `fcitx5`, and that `fcitx5` is running (`pgrep fcitx5`).
Works in some apps, not others	Install `fcitx5-frontend-gtk4` / `fcitx5-frontend-qt6`. For Flatpak apps: `flatpak override --user --env=GTK_IM_MODULE=fcitx`.
Candidate window in a screen corner (GNOME Wayland)	Install the kimpanel GNOME extension.
Chromium / Electron apps on Wayland	Start with `--enable-wayland-ime`, or run them under XWayland.
No `rq` dates; terms replaced not listed	`librime-plugin-lua` is missing; install it and run `./install.sh --no-apt`.
Changes to word lists have no effect	Tray menu → *Deploy*, or `fcitx5 -r`. Check `/tmp/hk-pinyin-deploy.log`.
A wrong word keeps coming first	Select the right one a few times; Rime learns. To forget learned words: `./uninstall.sh --purge` and reinstall.

8 Removing

```
./uninstall.sh          # remove schema and themes, keep learned words
./uninstall.sh --purge  # also delete learned words and your word list
im-config -n ibus       # optional: go back to IBus
```

9 Known limits

- `Caps Lock` types capital letters; it does not switch input sources as on macOS (Linux has no clean way to stop Caps Lock also toggling capitals). Use `Shift`.
- There is no cloud or AI prediction. Conversion uses Rime's statistical model and your history.
- How the not-yet-converted pinyin looks inside the text field (underlined or highlighted) depends on the application.
- Written-Cantonese characters are typed with approximate Mandarin readings, not Jyutping. For full Cantonese input, add Rime's `jyut6ping3` schema alongside.
- Tested with Ubuntu 24.04, fcitx5 5.1.7, librime 1.10, librime-lua 2023-09.

Built on fcitx5 (LGPL-2.1), librime and luna_pinyin (BSD-3-Clause/LGPL-3.0), librime-lua (BSD-3-Clause), OpenCC (Apache-2.0) and Noto CJK fonts (OFL-1.1).